OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

11.3 Lane sections

Lanes may be split into multiple lane sections. Each lane section contains a fixed number of lanes.

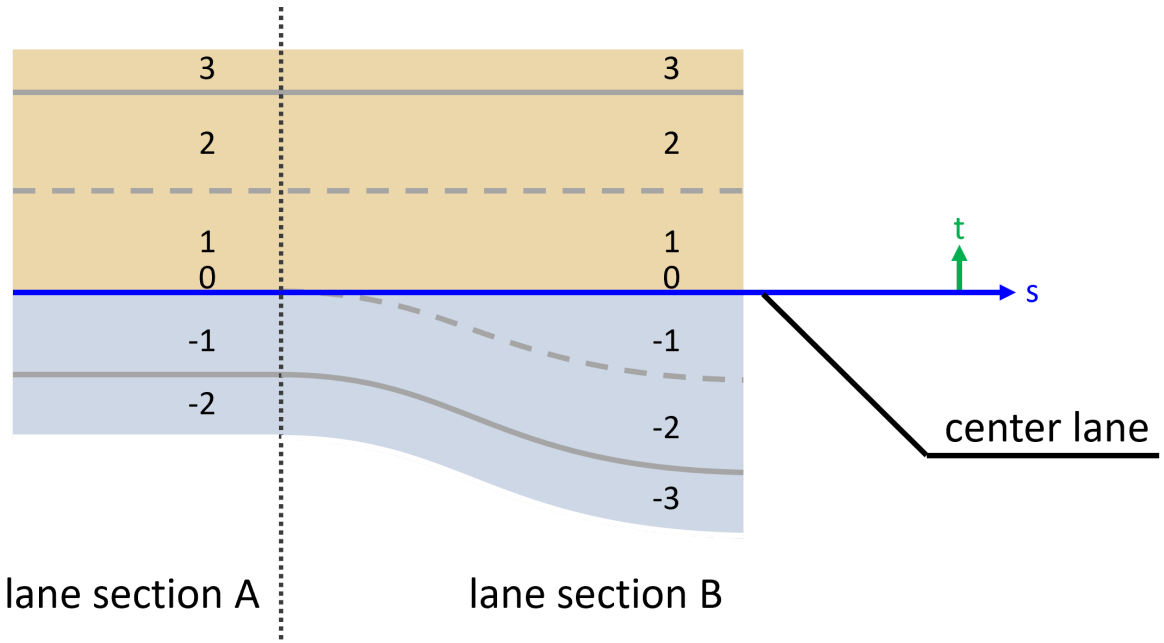

Figure 61. Road section with lane sections

Figure 61 shows that every time the number of lanes changes, a new lane section is required. Lane sections are defined in ascending order along the road reference line.

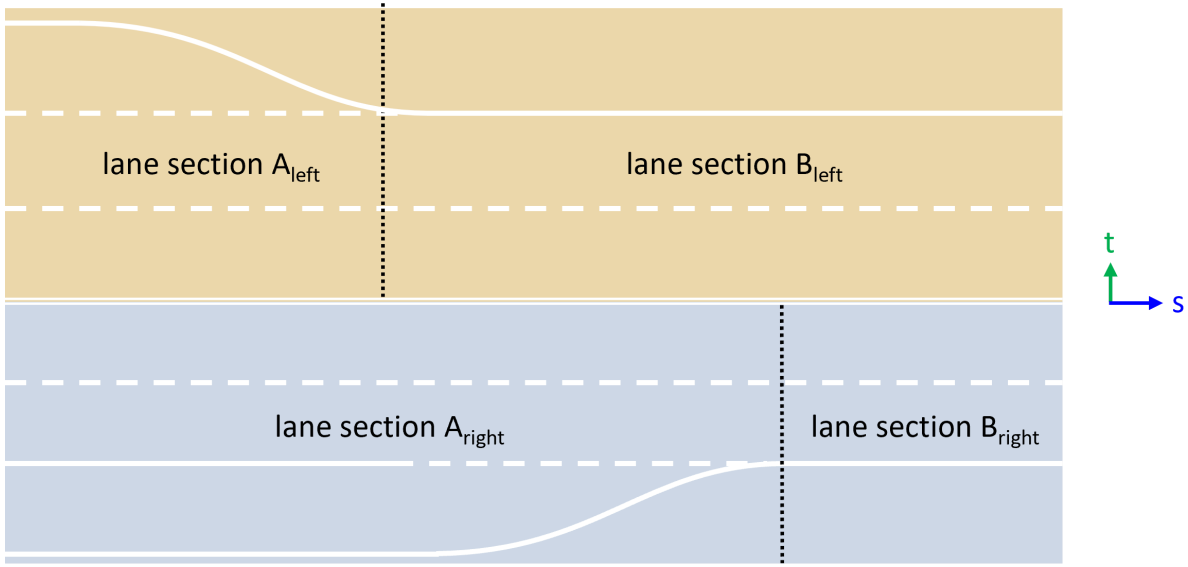

Figure 62. Lane sections defined separately for both sides of the road

Figure 62 shows how lane sections for complex roads may be defined for one side of the road only, using the @singleSide attribute.

Elements in UML model

`<laneSection>` element

In ASAM OpenDRIVE, lane sections are represented by `<laneSection>` elements within the `<lanes>` element.

```
UML class: t_road_lanes_laneSection
XML tag:   <laneSection> (Multiplicity: 1..*)
```

A lane section splits a road into multiple parts whenever the number of lanes or their function changes.

The distance between two succeeding lane sections shall not be zero.

For easier navigation through an ASAM OpenDRIVE road description, the lanes within a lane section are grouped into left, center, and right lanes. Each lane section shall contain one `<center>` element and at least one `<right>` or `<left>` element.

Table 37. Attributes of the <laneSection> element

Name	Type	Use	Unit	Description
`s`	t_grEqZero	required	m	s-coordinate of start position
`singleSide`	t_bool	optional		Lane section element is valid for one side only (left, center, or right), depending on the child elements.

For the child elements of the `<laneSection>` element refer to Lane groups.

Rules

The following rules apply to lane sections:

- asam.net:xodr:1.4.0:road.lane_section.lane_sect_req: Each road shall have at least one lane section.
- asam.net:xodr:1.4.0:road.lane_section.elem_asc_order: `<laneSection>` elements shall be defined in ascending order according to the s-coordinate.
- asam.net:xodr:1.4.0:road.lane_section.valid_length: The length of lane sections shall be greater than zero.
- There shall always be exactly one center lane at each s-position.
- Using lanes with a width of 0 for long distances should be avoided.
- asam.net:xodr:1.4.0:road.lane_section.lanesec_usage_lane_num: A new lane section shall be defined each time the number of lanes change.
- A lane section shall remain valid until a new lane section is defined.
- The properties of lanes inside a lane section may be changed as often as needed.
- Lane sections may be defined for one side of the road only using the @singleSide attribute.

Related topics

- Section 9.2, "Road reference line"
- Section 11.1, "Introduction to lanes"
- Section 11.2, "Lane groups"